Cardiovascular System (CARD) · NPLEX-P1-CARD-09

CV-9 Blood Distribution to Body Regions

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Space the two brain dumps for one competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-09-01-C01				
NPLEX-P1-CARD-09-01-C02				
NPLEX-P1-CARD-09-01-C03				
NPLEX-P1-CARD-09-01-C04				
NPLEX-P1-CARD-09-01-C05				
NPLEX-P1-CARD-09-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C01-BD1

Draw the aorta and label every branch with its level

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C01 Activity: NPLEX-P1-CARD-09-01-C01-BD1 Date: _____

Without looking at your notes, draw the aorta from the aortic valve to the bifurcation as one long vessel and label every branch in order. Start with the ascending aorta and write what it gives and the level where it ends. For the arch, name the three branches from right to left, write where the brachiocephalic trunk divides and into what, and mark the level the arch sits at. For the descending thoracic aorta write its upper and lower levels and list the branches it gives, including which intercostal spaces it supplies and where the first two come from instead. For the abdominal aorta write each branch with its vertebral level and what it supplies, in order from the diaphragm down to the bifurcation, and mark which branches are paired. Finish by writing the level at which the aorta passes the diaphragm and the level at which it divides.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C01-BD2

Three patients with a lesion on the aorta

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C01 Activity: NPLEX-P1-CARD-09-01-C01-BD2 Date: _____

Use the levels you just drew on three new patients. Patient 1 is a 54-year-old with a hoarse voice and a widened mediastinum: write which nerve is involved, what it is lying against, and why the aorta can injure it. Patient 2 has an aortic dissection repaired with a clamp on the descending thoracic aorta and wakes with weak legs, no pain or temperature sensation below the lesion, and intact vibration and position sense: write which artery was interrupted, which part of the cord lost its supply, and why vibration and position sense were spared. Patient 3 is a 28-year-old with high arm pressure, weak and delayed femoral pulses, and notching of the lower borders of ribs 3 to 8: write where the narrowing sits relative to the arch branches, why the arm pressure is high while the legs are underperfused, and why ribs 1 and 2 show no notching. For each patient write one line naming the vertebral level or landmark that places the lesion.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-09-01-C01-RT01

Rebuild the collateral route around a coarctation

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C01 Activity: NPLEX-P1-CARD-09-01-C01-RT01 Date: _____

Rebuild the six-step route blood takes around an adult coarctation, one step per line and in order, starting from the high pressure above the narrowing and ending with the change seen on the ribs. Draw the chest wall in outline and trace the route on your drawing, labelling the subclavian artery, the internal thoracic artery, the anterior and posterior intercostal arteries, and the descending aorta below the narrowing. Mark with an arrow the one vessel in which flow runs backwards compared with normal, and write one sentence on what drives it that way. Then write which ribs are notched and which two are spared, and name the trunk that supplies the spared spaces.

Rebuild from memory: Collateral route around an adult coarctation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C02-BD1

Draw the circle of Willis and its feeding vessels

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C02 Activity: NPLEX-P1-CARD-09-01-C02-BD1 Date: _____

Draw the base of the brain from below and build the circle of Willis from memory. Start with the two inflow systems: name the pair of vessels that forms the anterior circulation and the pair that forms the posterior circulation, and write where the second pair runs in the neck and what it joins to become. Label every vessel that forms the ring itself, then add the branches that leave it, including the artery that supplies the eye, the thin deep branches that leave the middle cerebral trunk at right angles, and the three cerebellar arteries with the vessel each arises from. Beside each labelled vessel write the territory it supplies. Mark the point where the oculomotor nerve emerges and the two vessels it lies between. Finish by writing why the external carotid artery supplies no brain tissue and naming the one external carotid branch that still matters inside the skull, with where it runs.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C02-BD2

Place four deficits on the vascular map

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C02 Activity: NPLEX-P1-CARD-09-01-C02-BD2 Date: _____

Four patients lose brain blood supply. Patient 1 has weakness of the contralateral leg more than the arm with new incontinence and a change in behavior. Patient 2 has vertigo, difficulty swallowing, hoarseness, a drooping eyelid and small pupil on one side with loss of pain and temperature sensation on that side of the face, and loss of pain and temperature sensation on the other side of the body. Patient 3 has a contralateral homonymous hemianopia with the central field spared. Patient 4 has a pure contralateral hemiparesis with no sensory or language loss and a long history of untreated hypertension. For each patient name the occluded artery, write which circulation it belongs to, and write one sentence tying the deficit to the territory. For patient 2 write what the combination of findings on opposite sides of the face and body tells you about the level of the lesion. For patient 4 write what chronic hypertension does to those particular vessels and the two other lesions it produces in them. Finish by writing why an embolus lodged in a cortical middle cerebral branch still infarcts tissue even when the circle of Willis is complete.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C03-BD1

Limb arteries, their landmarks and their pulse points

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C03 Activity: NPLEX-P1-CARD-09-01-C03-BD1 Date: _____

Write out the arterial trunk of the upper limb and then of the lower limb as a chain of named segments from the aorta to the hand and to the foot. Beside each segment write the landmark at which the name changes and, where a pulse can be felt, exactly where you would place your fingers and against what. Mark the two pulses that are often faint and the one that is absent in a small number of healthy adults. Then write where each trunk ends and name the vessels that rejoin in the hand and in the foot. Add the three nerve and artery pairs from the notes, naming the bony or soft tissue landmark each pair travels with. Finish by writing the order of the structures in the femoral triangle from lateral to medial.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C03-BD2

Localize four limb lesions from the examination

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C03 Activity: NPLEX-P1-CARD-09-01-C03-BD2 Date: _____

Work each of these from the pulse sequence and the landmarks. Patient 1 is a 66-year-old whose left femoral pulse is normal, popliteal pulse is faint, and dorsalis pedis and posterior tibial pulses are absent: write where the obstruction lies and which muscle group aches first on walking. Patient 2 has an ankle systolic pressure of 70 mm Hg with an arm systolic pressure of 140 mm Hg: calculate the ankle-brachial index, write what the normal relationship between ankle and arm pressure is and why, and state what this value means. Patient 3 is a child with a supracondylar humeral fracture whose radial pulse is absent and whose forearm is tense and painful: write which artery is at risk, which muscles are ischemic, and what the healed result is called. Patient 4 has a displaced intracapsular femoral neck fracture: write which artery supplies the femoral head and what the head is at risk of. Finish by writing where emboli most often lodge in the lower limb and why a block in the axillary artery above the subscapular origin is better tolerated than one below it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C04-BD1

Draw venous return and the portal system

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C04 Activity: NPLEX-P1-CARD-09-01-C04-BD1 Date: _____

Draw the two venae cavae and the hepatic portal system on one page. For the superior vena cava write which veins unite to form it, where they unite, and which vein joins it and at what level. For the inferior vena cava write the level at which it forms, which side of the aorta it ascends on, and the level at which it passes the diaphragm. Draw both renal veins and both gonadal veins and write where each gonadal vein drains, then mark the two structures the left renal vein passes between and write what else crosses in that same angle. For the portal system, name the two veins that unite to form the portal vein and where they meet, then write the share of cardiac output the liver receives, how much of that is portal, and how much of the liver's oxygen each source supplies. Write the normal portal pressure and the normal pressure in the hepatic veins, and note what the absence of valves in the portal system means for any rise in resistance. Finish by listing the four portosystemic anastomoses with the portal tributary and the systemic vein at each site.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C04-BD2

Four patients with an obstructed vein

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C04 Activity: NPLEX-P1-CARD-09-01-C04-BD2 Date: _____

Patient 1 has a mediastinal tumor, a swollen face, neck and arms, and distended veins over the chest wall. Write which vein is obstructed, then write the two routes blood can take back to the heart and which route is used when the block sits above the entry of the azygos vein and which when it sits below it, naming the vein in which flow reverses. Patient 2 is a 22-year-old man with flank pain, blood in the urine, and a new left-sided varicocele. Write which vein is compressed and between which two structures, and explain from the course of the two gonadal veins why varicocele is usually on the left. Patient 3 is a 58-year-old man with a new isolated right varicocele: write what that should make you look for and why. Patient 4 has cirrhosis with dilated veins radiating from the umbilicus, an enlarged spleen and a low platelet count. Write the site and the two veins involved in the umbilical finding, write why the spleen enlarges when the resistance lies in the liver, and write what the enlarged spleen does to the platelet count and why. Finish by writing the three sites at which raised portal pressure can originate, with one example for each, and the pressure gradient at which varices appear.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-09-01-C04-RT01

Rebuild the cycle that keeps portal pressure high

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C04 Activity: NPLEX-P1-CARD-09-01-C04-RT01 Date: _____

Rebuild the six-step cycle that holds portal pressure high in cirrhosis, one step per line and in order, then draw it as a closed loop with arrows so the last step feeds back into the first. Beside the first step write the three changes inside the liver that raise resistance. Beside the second step write which mediator dilates the splanchnic arterioles. Beside the fourth step write what the body is sensing when it activates the renin angiotensin aldosterone system and the sympathetic nerves, even though total blood volume is not low. Underneath the loop write the relationship between pressure, resistance and inflow in one line, and state which of the two cirrhosis raises. Finish by writing which step also produces ascites.

Rebuild from memory: Why portal pressure stays high in cirrhosis

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-09-01-C04-CA01

Vomiting blood in a man with long-standing liver disease

A 57-year-old man is brought to the emergency department after vomiting a large volume of bright red blood twice in one hour. He has had alcohol use disorder for 20 years and was told four years ago that he had cirrhosis. Temperature is 36.6 degrees C (97.9 degrees F), pulse is 116/min, respirations are 20/min, and blood pressure is 92/58 mm Hg. The abdomen is distended with shifting dullness, the spleen is palpable 4 cm below the left costal margin, and dilated veins radiate from the umbilicus. Platelet count is 68,000/mm3. Portal pressure measured during the admission gives a hepatic venous pressure gradient of 14 mm Hg.

1. Recognize. Name the vessels that bled, the portal tributary and the systemic vein that meet at that site, and the vein into which that systemic channel drains.

2. Explain. Explain why raised resistance inside the liver sends portal blood through the lower esophagus rather than across the liver to the inferior vena cava, and why the two other findings on this abdomen appear at the same time.

3. Differentiate. Write how this bleeding site differs from the anorectal anastomosis and from internal hemorrhoids, and what separates a bleed caused by raised portal pressure from one caused by a lesion in the stomach wall.

4. Investigate. Write which measurements and venous sites you would use to confirm where the raised resistance lies, and what value or finding each one would give in cirrhosis.

5. Decide. State where the pressure gradient in this patient sits relative to the normal gradient and the varix threshold, and name the site at which the resistance in this man is raised.

6. Justify. Justify that location from the pressures on either side of the liver, the absence of valves in the portal system, and the way the gradient behaves in prehepatic and posthepatic causes.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C05-BD1

Map the watershed zones and explain why they fail first

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C05 Activity: NPLEX-P1-CARD-09-01-C05-BD1 Date: _____

Define a watershed zone in one sentence, then explain in three or four sentences why pressure at that border is lower than anywhere else in either arterial tree and why the collateral channels there cannot rescue it. Write the range of mean arterial pressure over which cerebral blood flow stays nearly constant, what happens to flow below that range, and what chronic hypertension does to the range. Then draw a brain in lateral view and mark the cortical border zones, naming the two arteries that meet at each one and the region of the hemisphere each strip occupies, and mark the deep border zone. Draw the colon in outline and mark the two sites with weak marginal connections, naming the arteries that meet at each and writing which part of the large bowel has an extra supply and from where. Finish by listing the five regions that fail first in low flow, with the reason each one is vulnerable and the injury that results.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C05-BD2

Read a pattern of injury back to a mechanism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C05 Activity: NPLEX-P1-CARD-09-01-C05-BD2 Date: _____

A 71-year-old man is resuscitated after 50 minutes of severe hypotension during surgery. Over the next two days he passes bloody stools with left-sided abdominal pain, is found to have weakness of both shoulders and upper arms with normal grip strength, has rising aminotransferases, and has a rising creatinine with granular casts in the urine. For each of the four findings write the region injured, the arteries or the position on the supply path that makes it vulnerable, and the name of the injury. Write what the four findings have in common and the single mechanism that links them. Then write how this pattern differs from the pattern a cardiac embolus produces, covering the number of territories involved, whether findings are one-sided or two-sided, and the depth of bowel injury. Finish by writing why a man with a severe carotid stenosis can infarct a border zone after a fall in pressure that another patient would tolerate.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-09-01-C05-CP01

Global hypoperfusion against occlusion of one artery

Fill in every cell from memory, then check your table against the notes and correct what you missed.

Feature	Global hypoperfusion	Occlusion of a single artery
Mechanism of the fall in perfusion		
Where the infarct sits within an arterial territory		
One side or both sides		
Number of organs involved at once		
Shape of the cerebral lesion and the arm weakness it produces		
Depth of bowel wall injury		
Typical cause		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-09-01-C06-BD1

Build the flow and extraction table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C06 Activity: NPLEX-P1-CARD-09-01-C06-BD1 Date: _____

On blank paper build a table with one row for the gut and liver, the kidneys, resting muscle, the brain, the heart and the skin. Use the columns share of resting cardiac output, oxygen extraction, and what the flow is used for, and fill every cell. Rank the rows by share of output and then rank them again by extraction, and write one sentence on why the two rankings differ. Above the table write why each organ can set its own share of cardiac output while every organ sees the same aortic pressure. Below the table write the oxygen content of arterial blood per decilitre, the whole body figures for cardiac output, arteriovenous oxygen difference, oxygen consumption, extraction and mixed venous saturation, and the Fick principle as an equation. Finish with one line on what a low extraction tells you about an organ's reserve and one line on what a high extraction means for how that organ must meet extra demand.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-09-01-C06-BD2

Work the numbers, then explain two organs under strain

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-9 Blood Distribution to Body Regions
Competency: NPLEX-P1-CARD-09-01-C06 Activity: NPLEX-P1-CARD-09-01-C06-BD2 Date: _____

Arterial oxygen content is 20 mL/dL in every part of this activity. Kidney blood flow is 11 dL/min and renal venous content is 18.5 mL/dL. Coronary blood flow is 2.25 dL/min and coronary sinus content is 6 mL/dL. Calculate the arteriovenous oxygen difference, the oxygen consumption and the extraction for each organ, showing your working. Write which organ receives more blood and which uses more oxygen, and explain the mismatch from what each organ uses its flow for. Then let demand double in each organ and write how each one can meet it, giving the extraction the kidney would need and the extraction the heart would need, and say which figure is impossible and why. Write what the heart must do instead and name the two mediators that do it. Finish with the brain: write its share of body weight, its share of resting oxygen use, how long consciousness lasts without flow, and what rising arterial carbon dioxide and hyperventilation each do to its arterioles.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-09-01-C06-PR01

Predict the redistribution of cardiac output in hemorrhage

A 30-year-old man loses 1.5 L of blood from a stab wound to the thigh. Predict each item below, write the direction of the change, and give the receptor, nerve or local mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to arterial pressure and to baroreceptor firing		
What happens to sympathetic outflow and which receptor on the arterioles it acts through		
What happens to flow through the skin, the splanchnic bed, the kidneys and resting muscle		
What happens to the blood held in the splanchnic veins and what that does to venous return		
What happens to the diameter of cerebral and coronary arterioles while the other beds constrict		
What happens to the share of cardiac output reaching the brain and heart		
What happens to the oxygen extraction of the constricted beds, and what happens to those organs if the low flow continues		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-09-01-C01-Q01

Weight loss and vomiting with an aortic landmark

A 24-year-old woman comes to the physician because of 3 months of vomiting after meals and a 9 kg weight loss. She describes relief when she lies on her left side. Imaging shows dilatation of the first and second parts of the duodenum with narrowing of the third part where it crosses in front of the aorta, and a reduced angle between the aorta and the artery that passes in front of the duodenum. Which of the following arteries is responsible for the obstruction?

A. Celiac trunk at the level of the twelfth thoracic vertebra

B. Inferior mesenteric artery at the level of the third lumbar vertebra

C. Superior mesenteric artery at the level of the first lumbar vertebra

D. Right renal artery at the level of the second lumbar vertebra

Why did you pick it?

Board-style question NPLEX-P1-CARD-09-01-C02-Q02

Dizziness while working with one arm raised

A 64-year-old man who has smoked for 40 years becomes dizzy and unsteady whenever he paints a ceiling with his left arm. The left radial pulse is weak and delayed, and left arm systolic pressure is 30 mm Hg lower than the right. Neurologic examination between episodes is normal. Which of the following best explains his dizziness during use of the arm?

A. Reversal of flow down the left vertebral artery into the dilated arteries of the exercising arm

B. Embolization from an ulcerated plaque in the left internal carotid artery to the middle cerebral artery

C. Compression of the left subclavian artery between the anterior and middle scalene muscles by a cervical rib

D. Occlusion of the left posterior inferior cerebellar artery supplying the lateral medulla

Why did you pick it?

Board-style question NPLEX-P1-CARD-09-01-C05-Q03

Weak arms after a prolonged fall in blood pressure

A 68-year-old woman is resuscitated after a cardiac arrest lasting 40 minutes. On the second day she is awake and has weakness of both shoulders and both proximal arms, while grip strength, facial movement and leg strength are normal. Imaging shows narrow strips of infarction in both hemispheres along the upper convexity. Which of the following best explains the location of these infarcts?

A. Embolic occlusion of both anterior cerebral arteries beyond the anterior communicating artery

B. Thrombosis of the superior sagittal sinus with venous congestion of the parasagittal cortex

C. Hyaline arteriolosclerosis of the lenticulostriate arteries supplying the internal capsule

D. Perfusion pressure at the distal ends of the anterior and middle cerebral territories falling below the autoregulatory range

Why did you pick it?